

Side Topic: Selection vs. Treatment

ECON 5001
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Selection vs. Treatment

Example: OSU students were only required to live on campus for one year. Then:

“OSU students who choose to live on campus their 2nd year
are more likely to have a good GPA.”

Competing Explanations:

Selection vs. Treatment

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 - Better study habits?
 - Fewer distractions?

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Competing Explanations:

1. Living on campus *causes* higher GPAs
 - Better study habits?
 - Fewer distractions?
2. Only nerds choose to live on campus a 2nd year!!

Selection vs. Treatment

Imagine that “living on campus” is like a drug trial.

- **Treatment:** The drug treatment actually works. “Living on campus” actually causes better GPAs.
- **Selection:** The drug treatment doesn’t do anything! It’s just that those who select into the drug trial are different than those that don’t.

With this data alone, we can’t know which is true (or both)

Selection vs. Treatment

More examples:

- **Headline:** People who learn debate skills are more likely to have leadership roles.
 - **Treatment:** Debate helps you learn leadership skills.
 - **Selection:** People who choose debate already have good leadership skills
- **Headline:** People who drink in moderation are more likely to live longer.
 - **Treatment:** Alcohol harms your health
 - **Selection:** Homebodies don't drink much
- **Headline:** People who live together before marriage are more likely to divorce.
 - **Treatment:** Cohabitation harms relationship/commitment dynamics
 - **Selection:** People who live together are less religious/more “okay” with divorce

Selection vs. Treatment

“Selection” doesn’t have to be voluntary.

- **Headline:** People who are from disadvantaged backgrounds are more likely to have an undiagnosed eating disorder.
 - **Treatment:** Poor $\Rightarrow$ stress $\Rightarrow$ eating disorder
 - **Selection:**
- **Headline:** People who use PTO at work are more likely to have high job satisfaction.
 - **Treatment:** PTO $\uparrow$ job satisfaction
 - **Selection:** If you have a great job, you have both PTO and satisfaction

Selection vs. Treatment

Getting political:

- **Headline:** People who vote for “my party” are more likely to [do some really great thing].
- **Headline:** People who vote for “the other party” are more likely to [be horrible in some way].

Be very critical!

1. Is that correlation even true?
2. There are **tons** of possible selection effects:
 - Education
 - Race
 - Age
 - Religion
 - ...

...and that makes it easy to find crazy correlations

What's the Solution

What are the solutions?

1. **Random assignment!** Randomly assign 1/2 of OSU 2nd years to the dorms, 1/2 to off-campus apartments.
Selection effects are impossible! Only treatment remains.
2. If random assignment is impossible (eg, being poor), just be honest about the uncertainty.

Growing distrust in both researchers and media...

- Read and decide for yourself